

7.6 Special Right Triangles

Lesson Introduction

 Benchmark	MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Targets	- I can use side ratios of 45–45–90 triangles to solve problems. - I can use side ratios of 30–60–90 triangles to solve problems.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.			
 Essential Questions	- How can I use the side ratios of 45-45-90 triangles to solve problems? - How can I use the side ratios of 30-60-90 triangles to solve problems?			
 Lesson Narrative	In this lesson, students are introduced to the 45-45-90 and 30-60-90 special right triangles. Through exploration, students will discover the side relationships in special right triangles. They will then be formally introduced to 45-45-90 and 30-60-90 triangle formulas to find side lengths. Students will use this knowledge to solve problems involving these special right triangles.			
 Component Summary	7.6.1 Warm-Up (~5 min.)	Determine the shaded amount (<i>SE p. 400</i>)	7.6.3 Guided Instruction (5-10 min.)	Formalize the 45-45-90 and 30-60-90 triangle side relationships (<i>SE p. 405</i>)
	7.6.2 Exploration (20-25 min.)	Discover the side relationships in special right triangles (<i>SE p. 401</i>)	7.6.4 Guided Practice (5 min.)	Determine side measures in special right triangles (<i>SE p. 406</i>)
	7.6.5 Wrap-Up (~5 min.)	Demonstrate understanding of using special right triangles (<i>SE p. 407</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		N/A	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not recognize when a triangle meets the conditions of a special right triangle. - Students may apply the side relationships to the wrong special right triangles or other triangles.	- This lesson has fewer activities than some of the other lessons in this unit. Allow students ample time to engage in the 7.6.2 Exploration for students to discover the special triangles without giving too much up-front information. Students will see the formalized relationships in 7.6.3 Guided Instruction. - If time permits at the conclusion of the lesson, consider showing students a side-by-side comparison of using the side relationships in special right triangles and solving when the relationships are not used.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 7.6.2 Exploration provides students with the opportunity to discover special right triangles. During this activity, students will have quiet time to think about the problems and work on them individually, and then time to share their response or their progress with a partner, and later the whole group.	MA.K12.MTR.6.1: Reasonableness This lesson requires students to perform multiple steps to find missing sides of special right triangles. Due to the problems requiring students to use order of operations, students may make minor errors that result in an incorrect solution. Encourage students to always check each step of their work and ask students to verify their solutions by explaining their work.
 Differentiate Instruction	Consider creating a side-by-side visual for determining side lengths using the side relationships of special right triangles versus not using the special right triangle side relationships, and the process for solving problems in both scenarios. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

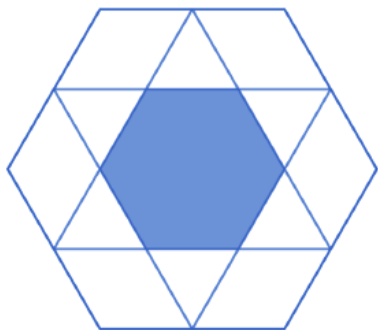
7.6.1 Warm-Up: Critical Thinking

Component Overview: Students will review the shading in a figure composed of multiple shapes in order to determine the fraction of the shape that is shaded.



7.6.1 Warm-Up: Critical Thinking

1. What fraction of the shape is shaded? $\frac{1}{4}$



2. Describe your problem-solving method.

Answers will vary.

Decompose the inner blue hexagon into congruent 6 congruent triangles. Then, decompose each parallelogram on the outer edges into 2 congruent triangles (6 parallelograms $\times$ 2 triangles). Count the total number of triangles that are in the entire large hexagon (24 triangles). 6 out of 24 triangles are $\frac{1}{4}$ of the hexagon.



Consider leveraging a Think-Pair-Share instructional routine with this activity. Allow students individual processing time to think through the problem on their own, and then work with a partner to discuss their answers and thinking around it to come to a consensus. Then have students share their thinking with the whole group.

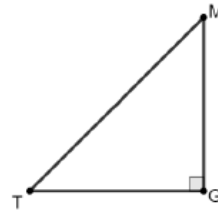
7.6.2 Exploration: Exploring Special Right Triangles

Component Overview: Students will explore 45-45-90 and 30-60-90 special right triangles to discover the side relationships.



7.6.2 Exploration: Exploring Special Right Triangles

1. $\triangle MGT$ is a right triangle, where $m\angle MGT = 90^\circ$, $MG = 6$, and $GT = 6$.



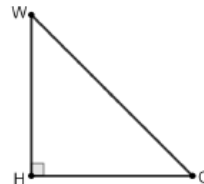
- a. What type of right triangle is $\triangle MGT$?

An isosceles right triangle

- b. Determine the measurements. Leave the measurement of $\overline{MT}$ in radical form.

MT	$m\angle GMT$	$m\angle GTM$
$(MG)^2 + (GT)^2 = (MT)^2$ $6^2 + 6^2 = (MT)^2$ $MT = 6\sqrt{2}$	$\tan^{-1}\left(\frac{6}{6}\right)$ 45°	$\sin^{-1}\left(\frac{6}{6\sqrt{2}}\right)$ 45°

2. $\triangle WCH$ is a right triangle, where $m\angle WHC = 90^\circ$, $m\angle WCH = 45^\circ$, and $WC = 4\sqrt{2}$.



- a. What type of right triangle is $\triangle WCH$?

An isosceles right triangle

- b. Determine the measurements of each.

HC	WH	$m\angle CWH$
$\cos(45^\circ) = \frac{HC}{4\sqrt{2}}$ $HC = 4$	$\sin(45^\circ) = \frac{WH}{4\sqrt{2}}$ $WH = 4$	$m\angle CWH + 45^\circ + 90^\circ = 180^\circ$ 45°



This exploration is scaffolded to allow students to discover the relationships of the sides of special right triangles with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and share your thinking out loud in the next component.



Students may provide different correct answers for question 1a. They may write answers such as “a right triangle.” Display all correct responses on the board until the most precise description of “isosceles right triangle” is determined.



If students get an error message when inputting the expression for $m\angle GTM$ in the calculator, they most likely did not group the denominator when dividing. Ask clarifying questions that lead to the understanding of using the grouping symbol when inputting.



If students answer $HC = \frac{1}{8}$ and/or $WH = \frac{1}{8}$, then they either set up the equation incorrectly, or solved the correct equation incorrectly.

7.6.2 Exploration: Exploring Special Right Triangles (continued)

3. What do you notice about the angles of $\triangle WCH$ and $\triangle MGT$?

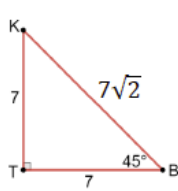
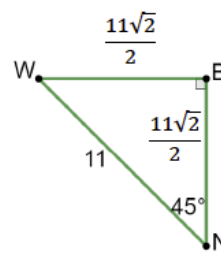
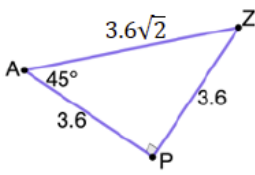
Answers will vary. They both have 45 – 45 – 90 degree angles.

4. What do you notice about the legs of $\triangle WCH$ and $\triangle MGT$?

*Answers will vary.
They both have two legs that are congruent and a hypotenuse that is the product of the leg length times $\sqrt{2}$.*

$\triangle WCH$ and $\triangle MGT$ are 45 – 45 – 90 triangles.

5. The table shows three 45 – 45 – 90 triangles.

$\triangle KBT$	$\triangle WBN$	$\triangle AZP$
		

- a. Discuss with your partner the relationship between the legs of a 45 – 45 – 90 triangle and its hypotenuse. Write a conjecture.

The hypotenuse is equal to the product of the length of a leg and $\sqrt{2}$.



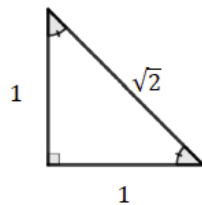
Consider converting questions 3 and 4 to a partner discussion. Students are more likely to share at-length verbally than write in their books. Have them notate their summary in their books for future reference.

7.6.2 Exploration: Exploring Special Right Triangles (continued)

- b. Ask two other pairs of partners for the conjecture they wrote. Listen to their conjecture and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

Conjecture	Initials
<i>Answers will vary.</i>	

6. An isosceles right triangle is shown.



Describe how the triangle shows the relationships in a 45 – 45 – 90 triangle.

The legs have the same measurement and the hypotenuse is the leg measurement times $\sqrt{2}$.



Have students try to find other groups with different opinions and approaches to question 5b.



Have students share out their descriptions on question 6 in a whole-class discussion. This is where students may have misconceptions that can be addressed during the discussion.

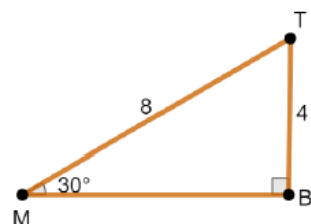
7.6.2 Exploration: Exploring Special Right Triangles (continued)

7. $\triangle MBT$ is a right triangle, where $m\angle MBT = 90^\circ$, $m\angle TMB = 30^\circ$, $TB = 4$, and $MT = 8$.

Find the length of $\overline{MB}$ using the Pythagorean Theorem. Write the answer in simplified radical form.

$$4^2 + MB^2 = 8^2$$

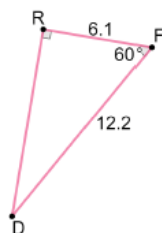
$$MB = 4\sqrt{3}$$



8. $\triangle RFD$ is a right triangle, where $m\angle DRF = 90^\circ$, $m\angle RFD = 60^\circ$, $RF = 6.1$, and $FD = 12.2$.

Discuss with your partner what features $\triangle RFD$ has in common with $\triangle MBT$.

All angle measures are the same and the leg across from the 30° angle is half the length of the hypotenuse.



9. $\triangle MBT$ and $\triangle RFD$ are 30 – 60 – 90 triangles.

The table gives the exact measurements for each of the sides of $\triangle MBT$ and $\triangle RFD$.

Triangle	Short Leg	Long Leg	Hypotenuse
$\triangle MBT$	4	$4\sqrt{3}$	8
$\triangle RFD$	6.1	$6.1\sqrt{3}$	12.2

Complete the table by describing the relationship between the side lengths of a 30 – 60 – 90 triangle.

Sides	Relationship
short leg and hypotenuse	<i>The hypotenuse is double the short leg.</i>
short leg and long leg	<i>The long leg is the product of the short leg times $\sqrt{3}$.</i>



Consider alternative ways of assessing students' understanding by having students explain their thought process for question 7:

- How did you determine the length of $\overline{MB}$?



If students struggle to determine the relationship, provide a different entry point by having the students label the measurement of all missing sides and angles prior to stating what the two triangles have in common.



- Why is knowing that the two triangles are 30-60-90 triangles beneficial?
- How are the sides of one 30-60-90 triangle related to the sides of another 30-60-90 triangle?

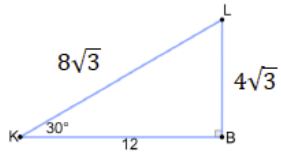
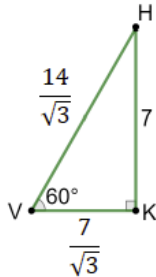
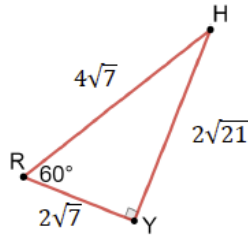
Listen for the correct usage of the vocabulary “similar triangles” and “proportional sides.”



For the table, listen to student descriptions of these relationships for appropriate academic language and specific relationships. Students may list relationships that are accurate but do not connect to the learning outcomes of this lesson.

7.6.2 Exploration: Exploring Special Right Triangles (continued)

10. The table below includes three 30 – 60 – 90 triangles.

$\triangle KBL$	
$\triangle VHK$	
$\triangle HRY$	



Students may not spend time evaluating these images as there is no “work” required to complete the table. Consider implementing a Notice and Wonder for these images so that students pause here before moving to the discussion part of the next question.

7.6.2 Exploration: Exploring Special Right Triangles (continued)

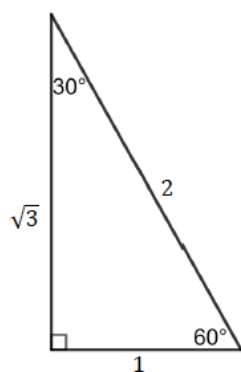
Discuss with your partner the relationships between the sides of a $30-60-90$ triangle in the table. Complete the table by showing how the short side of each of the triangles can be used to determine the long side and the hypotenuse. Write the final length as an exact value.

Triangle	Short Side	Work to Find the Long Side	Work to Find the Hypotenuse
$\triangle KBL$	$4\sqrt{3}$	$\frac{4\sqrt{3} \cdot \sqrt{3}}{4 \cdot 3}$ 12	$\frac{4\sqrt{3} \cdot 2}{8\sqrt{3}}$
$\triangle VHK$	$\frac{7}{\sqrt{3}}$	$\frac{\frac{7}{\sqrt{3}} \cdot \sqrt{3}}{7}$	$\frac{\frac{7}{\sqrt{3}} \cdot 2}{\frac{14}{\sqrt{3}}}$
$\triangle HRY$	$2\sqrt{7}$	$\frac{2\sqrt{7} \cdot \sqrt{3}}{2\sqrt{21}}$	$\frac{2\sqrt{7} \cdot 2}{4\sqrt{7}}$

11. Describe how the triangle shows the relationships in a $30-60-90$ triangle.

The hypotenuse is twice the length of the short side.

The long side is the product of the short side times $\sqrt{3}$.



Consider having students switch partners and discuss their responses to question 10. Students can then return to their original partner and discuss any difference and make corrections.



Listen for student responses to question 11, and their rationale for why this triangle shows the relationships in $30-60-90$ triangles. Some students may use surface-level responses instead of using what aligns with the learning outcomes of this activity.

7.6.3 Guided Instruction: 45-45-90 and 30-60-90 Triangles

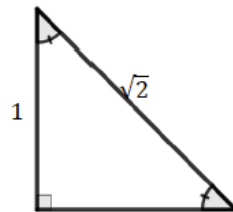
Component Overview: Students will be formally introduced to the side relationships of 45-45-90 and 30-60-90 special triangles.



7.6.3 Guided Instruction: 45-45-90 and 30-60-90 Triangles



The side lengths of a 45 – 45 – 90 triangle can be found using the equation $\text{leg} \cdot \sqrt{2} = \text{hypotenuse}$.



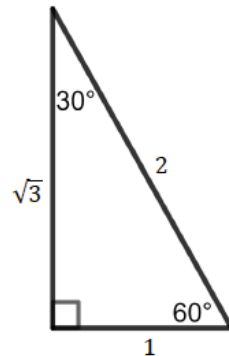
Students may write the exact form of question 1 as $\frac{12}{\sqrt{2}}$ or as $6\sqrt{2}$. Discuss why these are equivalent forms.

1. $\triangle ZYN$ has $m\angle YZN = m\angle YNZ = 45^\circ$, $m\angle ZYN = 90^\circ$, and $ZN = 12$. What are the lengths of the legs of the triangle? Leave your answer as an exact answer.

$$\begin{aligned} \text{leg} \cdot \sqrt{2} &= 12 \\ \text{leg} &= \frac{12}{\sqrt{2}} \end{aligned}$$



The lengths of the short leg and the long leg of a 30 – 60 – 90 triangle can be found using the equation $\text{short leg} \cdot \sqrt{3} = \text{long leg}$.



The lengths of the short leg and the hypotenuse of a 30 – 60 – 90 triangle can be found using the equation $\text{short leg} \cdot 2 = \text{hypotenuse}$.



Why do you think the process for finding the side lengths is different for the two legs of a 30-60-90 special right triangle?

7.6.3 Guided Instruction: 45-45-90 and 30-60-90 Triangles (continued)

2. $\triangle BHK$ is a 30 – 60 – 90 triangle with $m\angle KBH = 30^\circ$, $m\angle BKH = 90^\circ$, and $BK = 5\sqrt{3}$. What are the measures of the other sides of the triangle?

Short Side

$$KH \cdot \sqrt{3} = 5\sqrt{3}$$

$$KH = 5$$

Hypotenuse

$$KH \cdot 2 = BH \quad 10 = BH$$



Encourage students to create a visual representation of this problem to help them see the angles and sides in a way that is consistent with what they have experienced in this and prior activities.

7.6.4 Guided Practice: Solving Special Right Triangles

Component Overview: Students will complete two practice questions with solving special right triangles.



7.6.4 Guided Practice: Solving Special Right Triangles

1. $\triangle LPM$ is a $45 - 45 - 90$ triangle with $m\angle LMP = m\angle MLP = 45^\circ$, $m\angle LPM = 90^\circ$, and $LM = 7$. What are the lengths of $\overline{PM}$ and $\overline{LP}$?

$$\text{leg} \cdot \sqrt{2} = 7$$

$$PM \text{ and } LP = \frac{7}{\sqrt{2}}$$

2. $\triangle TNW$ is a right triangle with $m\angle NTW = 60^\circ$, $m\angle WNT = 90^\circ$, and $TW = 18$. What are the measures of the other sides of the triangle?

$$NT \cdot 2 = 18$$

$$NT = 9$$

$$NT \cdot \sqrt{3} = WN$$

$$9\sqrt{3} = WN$$



Consider pulling together a small group of students who may need additional support based on your observations during the prior activities. Consider working with students on question 1 in small groups and releasing them to try question 2 on their own or in a small group.

7.6.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door on solving a problem involving a special triangle. Students should not use a calculator to complete the wrap-up.



7.6.5 Wrap-Up: Ticket Out the Door

1. $\triangle FHX$ has angle measures $m\angle HXF = 90^\circ$ and $m\angle HFX = 30^\circ$, and side length $HX = 6\sqrt{3}$. What are the exact measures of $\overline{FX}$ and $\overline{FH}$?

$$6\sqrt{3} \cdot \sqrt{3} = FX$$

$$18 = FX$$

$$6\sqrt{3} \cdot 2 = FH$$

$$12\sqrt{3} = FH$$



Student responses to this activity can provide data to which components of the lesson, if any, may need additional review.